

Optical Fiber — Complete Formula Sheet

TOPIC 1: Numerical Aperture (NA)

1.1 Critical Angle Condition

$$\sin \phi_c = \frac{n_2}{n_1}$$

Parameter	Meaning
ϕ_c	Critical angle at core-cladding boundary
n_1	Refractive index of core
n_2	Refractive index of cladding

1.2 Snell's Law at Entry Face

$$n_0 \sin \theta_1 = n_1 \sin \theta_2$$

Parameter	Meaning
n_0	Refractive index of outside medium (air = 1)
θ_1	Angle of incidence at entry face
θ_2	Angle of refraction inside core

1.3 Geometric Relation (Key Step)

$$\theta_2 + \phi = 90^\circ \Rightarrow \sin \theta_2 = \cos \phi = \sqrt{1 - \sin^2 \phi}$$

▮ θ_2 and ϕ are complementary angles inside the fiber

1.4 Combined Entry + Critical Angle

$$n_0 \sin \theta_1 = n_1 \sqrt{1 - \sin^2 \phi}$$

1.5 Numerical Aperture — Exact Formula (Eq. 2.7 & 2.8)

$$NA = \sqrt{n_1^2 - n_2^2} = n_0 \sin \theta_a$$

Parameter	Meaning
NA	Numerical Aperture (light-gathering ability)
n1	Core refractive index
n2	Cladding refractive index
n0	Outside medium index (air = 1)
θa	Acceptance angle (half-angle of acceptance cone)

TOPIC 2: Fractional Refractive Index Difference (Δ)

2.1 Definition

$$\Delta = \frac{n_1 - n_2}{n_1} \quad (\text{approximate})$$

$$\Delta = \frac{n_1^2 - n_2^2}{2n_1^2} \quad (\text{exact, Eq. 2.9})$$

Parameter	Meaning
Δ	Fractional refractive index difference
n1	Core refractive index
n2	Cladding refractive index

2.2 NA in Terms of Δ (Approximate, valid when Δ ≪ 1)

$$NA \approx n_1 \sqrt{2\Delta}$$

Parameter	Meaning
NA	Numerical Aperture
n1	Core refractive index
Δ	Fractional index difference (typically ~1%)

Valid under **weakly guiding approximation** ($n_1 \approx n_2$)

2.3 Useful Rearrangement

$$n_2^2 = n_1^2 - 2\Delta n_1^2$$

TOPIC 3: Normalized Frequency (V-Number)

3.1 V-Number

$$V = \frac{2\pi a}{\lambda} \sqrt{n_1^2 - n_2^2} = \frac{2\pi a}{\lambda} \cdot NA$$

Parameter	Meaning
V	Normalized frequency (V-number)
a	Core radius
λ	Operating wavelength
n1, n2	Core and cladding indices

Single-mode condition: **V < 2.405**

3.2 Cutoff Wavelength (Single-Mode Fiber, Eq. 2.98)

$$\lambda_c = \frac{2\pi a n_1 (2\Delta)^{1/2}}{2.405}$$

Parameter	Meaning
λ_c	Cutoff wavelength
a	Core radius
n_1	Core refractive index
Δ	Fractional index difference
2.405	First zero of Bessel function J_0 (fixed constant)

Single-mode operation only occurs for $\lambda > \lambda_c$

TOPIC 4: Fiber Bend Losses — Critical Radius of Curvature

4.1 Multimode Fiber Critical Radius (Eq. 3.8)

$$R_c \approx \frac{3n_1^2\lambda}{4\pi(n_1^2 - n_2^2)^{3/2}}$$

Parameter	Meaning
R_c	Critical radius of curvature
n_1	Core refractive index
n_2	Cladding refractive index
λ	Operating wavelength

4.2 Single-Mode Fiber Critical Radius (Eq. 3.9)

$$R_{cs} \approx \frac{20\lambda}{(n_1^2 - n_2^2)^{3/2}} \left(2.748 - \frac{0.996\lambda}{\lambda_c} \right)^{-3}$$

Parameter	Meaning
Rcs	Critical radius for single-mode fiber
λ	Operating wavelength
n1, n2	Core and cladding indices
λ_c	Cutoff wavelength
2.748	Empirical constant (from Bessel function analysis)
0.996	Empirical constant ≈ 1 (from Bessel function analysis)

The term $(2.748 - 0.996\lambda/\lambda_c) \rightarrow$ approaches 0 as $\lambda \rightarrow \lambda_c \rightarrow R_c \rightarrow \infty$ (catastrophic loss!)

TOPIC 5: Rayleigh Scattering

5.1 Rayleigh Scattering Coefficient (Eq. 3.4)

$$\gamma_R = \frac{8\pi^3 n^8 p^2 \beta_c K T_F}{3\lambda^4}$$

Parameter	Meaning
γ_R	Rayleigh scattering coefficient (m^{-1})
n	Refractive index of material
p	Photoelastic coefficient
β_c	Isothermal compressibility ($\text{m}^2 \text{N}^{-1}$)
K	Boltzmann's constant = $1.381 \times 10^{-23} \text{ J K}^{-1}$
TF	Fictive temperature (K)
λ	Wavelength (m)

Most important relationship: $\gamma_R \propto 1/\lambda^4 \rightarrow$ shorter wavelength = much more loss

5.2 Transmission Loss Factor (Eq. 3.5)

$$\mathcal{L}_{km} = \exp(-\gamma_R L)$$

Parameter	Meaning
$\mathcal{L}_{km}$	Transmission loss factor (dimensionless, 0 to 1)
γ_R	Rayleigh scattering coefficient (m^{-1})
L	Length of fiber in meters (1 km = 10^3 m)

Always convert L to meters since γ_R is in m^{-1}

5.3 Attenuation in dB (Eq. 3.1)

$$\text{Attenuation} = 10 \log_{10} \left(\frac{1}{\mathcal{L}_{km}} \right) \text{ dB km}^{-1}$$

TOPIC 6: Fiber Attenuation & Power Budgeting

6.1 Power in dBm

$$P(\text{dBm}) = 10 \log_{10} \left(\frac{P(\text{mW})}{1 \text{ mW}} \right)$$

Value	dBm
$P = 1 \text{ mW}$	0 dBm
$P > 1 \text{ mW}$	positive dBm
$P < 1 \text{ mW}$	negative dBm

6.2 Convert dBm back to mW

$$P(\text{mW}) = 10^{\frac{P(\text{dBm})}{10}} \times 1 \text{ mW}$$

6.3 Total Allowable Loss

$$\alpha_{tot} = P_{in}(\text{dBm}) - P_{out}(\text{dBm})$$

Parameter	Meaning
α_{tot}	Total loss the signal can afford (dB)
P_{in}	Launched power (dBm)
P_{out}	Minimum required receiver power (dBm)

6.4 Maximum Link Length

$$L_{max} = \frac{\alpha_{tot} \text{ (dB)}}{\text{Attenuation (dB/km)}}$$

6.5 Minimum Launch Power (Power Budgeting)

$$P_{in}(\text{dBm}) = P_{out}(\text{dBm}) + \text{Total System Loss (dB)}$$

6.6 Total System Loss

$$\text{Total Loss} = \underbrace{(L \times \alpha)}_{\text{Fiber Loss}} + \underbrace{(N_j \times \alpha_j)}_{\text{Joint Loss}}$$

Parameter	Meaning
L	Fiber length (km)
α	Attenuation per km (dB/km)
N_j	Number of joints/splices
α_j	Loss per joint (dB)

#	Formula	Topic
1	$NA = \sqrt{(n_1^2 - n_2^2)}$	Numerical Aperture
2	$NA = n_0 \sin \theta_a$	Acceptance angle
3	$NA \approx n_1 \sqrt{(2\Delta)}$	NA via Δ (approx)
4	$\Delta = (n_1 - n_2) / n_1$	Fractional index diff
5	$V = (2\pi a / \lambda) \cdot NA$	V-number
6	$\lambda_c = 2\pi a n_1 \sqrt{(2\Delta)} / 2.405$	Cutoff wavelength
7	$R_c = 3n_1^2 \lambda / 4\pi (n_1^2 - n_2^2)^{3/2}$	Multimode bend radius
8	$R_{cs} = [20\lambda / (n_1^2 - n_2^2)^{3/2}] \cdot (2.748 - 0.996\lambda / \lambda_c)^{-3}$	Single-mode bend radius
9	$\gamma_R = 8\pi^3 n^s p^2 \beta_c KTF / 3\lambda^4$	Rayleigh coefficient
10	$\mathcal{L}_{km} = \exp(-\gamma_R \cdot L)$	Transmission factor
11	$Atten = 10 \log(1 / \mathcal{L}_{km})$	Attenuation dB/km
12	$P(\text{dBm}) = 10 \log(P / 1\text{mW})$	Power conversion
13	$L_{max} = \alpha_{tot} / \text{attenuation}$	Max link length
14	$P_{in} = P_{out} + \text{Total Loss}$	Min launch power